Cloud Computing

Lab 7

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Section: F

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1. Aim

To implement a distributed system using the Raft consensus algorithm in Docker containers and observe leader election, follower behavior, and fault tolerance.

2. Objectives

- Understand distributed systems
- Study leader election in Raft
- Simulate nodes using Docker
- Observe fault tolerance
- Understand Docker networking

3. Software Requirements

Docker, Ubuntu Terminal, Python (inside container)

4. Project Setup

A project folder named 'raft-docker-lab' was created. Files: node.py, Dockerfile, requirements.txt, docker-compose.yml.

```
PES1UG23AM343:/mnt/c/Users/acer/OneDrive/Desktop/raft-docker
-lab$ touch node.py Dockerfile requirements.txt docker-compo
se.yml
ls
Dockerfile  docker-compose.yml  node.py  requirements.txt
PES1UG23AM343:/mnt/c/Users/acer/OneDrive/Desktop/raft-docker
-lab$
```

5. Configuration

requirements.txt includes 'raftos'. Dockerfile sets Python environment. docker-compose.yml defines node1, node2, node3. node.py handles Raft logic.

6. Execution

Command:

sudo docker compose up --build

```
PES1UG23AM343:/mnt/c/Users/acer/OneDrive/Desktop/raft-docker-lab$ sudo docker compose up --build
[sudo] password for vardha:
[+] Building 119.8s (20/20) FINISHED
 => [internal] load local bake definitions                                                                          0.0s
 => => reading from stdin 1.52kB                                                                                    0.0s
 => [node3 internal] load build definition from Dockerfile                                                          0.1s
 => => transferring dockerfile: 489B                                                                                0.0s
 => [node1 internal] load metadata for docker.io/library/python:3.7.16-bullseye                                    6.0s
 => [node1 internal] load .dockerignore                                                                             0.1s
 => => transferring context: 2B                                                                                     0.0s
 => [node1 internal] load build context                                                                             0.1s
 => => transferring context: 1.41kB                                                                                 0.0s
 => [node1 1/9] FROM docker.io/library/python:3.7.16-bullseye@sha256:d98fd1fa8fd1756bfe883b5004a60cfc428a1ce6e575652b771b66c8db222478  57.5s
 => => resolve docker.io/library/python:3.7.16-bullseye@sha256:d98fd1fa8fd1756bfe883b5004a60cfc428a1ce6e575652b771b66c8db222478  0.1s
 => => sha256:f4b2de90806a5f69e4478b1658c6b90182c8eeffc7661b80fb6d0237f8a9b9c5 243B / 243B                         1.0s
 => => sha256:c1e5026c64577dee4b6bac20b18196f964a41d8b9016fbbbada0c70b601cd5bf 196.85MB / 196.85MB                38.4s
 => => sha256:f0016544b8b99a7e91472b231a8eb1af0aede28806efc821c60ca20b65fc1eb8 6.29MB / 6.29MB                     7.3s
 => => sha256:2d1440091f03b98cf78ad49ec8c4dd9fb6eff72f7e831a9980dfb7d785be7207 2.91MB / 2.91MB                     5.1s
 => => sha256:4226d6e6e614b0bcac20128496b229d45ca951e30ed0103cde290b5e3a263c44 15.04MB / 15.04MB                   9.7s
 => => sha256:bd73737482dd5575526c7207872963479808d979ab2741c321706b8553918474 55.05MB / 55.05MB                  23.0s
 => => sha256:75256935197ed1bb3b994a77c01efa00349b901014448a260fafd9c3719a741d 54.58MB / 54.58MB                  20.1s
 => => sha256:6710592d62aa1338ac1c1c363dedc255659f666cc41441c7e0f735c484db10ff 15.76MB / 15.76MB                  11.4s
 => => extracting sha256:bd73737482dd5575526c7207872963479808d979ab2741c321706b8553918474                          8.8s
 => => extracting sha256:6710592d62aa1338ac1c1c363dedc255659f666cc41441c7e0f735c484db10ff                          1.6s
 => => extracting sha256:75256935197ed1bb3b994a77c01efa00349b901014448a260fafd9c3719a741d                          3.5s
 => => extracting sha256:c1e5026c64577dee4b6bac20b18196f964a41d8b9016fbbbada0c70b601cd5bf                         12.4s
 => => extracting sha256:f0016544b8b99a7e91472b231a8eb1af0aede28806efc821c60ca20b65fc1eb8                          0.7s
 => => extracting sha256:4226d6e6e614b0bcac20128496b229d45ca951e30ed0103cde290b5e3a263c44                          1.2s
 => => extracting sha256:f4b2de90806a5f69e4478b1658c6b90182c8eeffc7661b80fb6d0237f8a9b9c5                          0.1s
 => => extracting sha256:2d1440091f03b98cf78ad49ec8c4dd9fb6eff72f7e831a9980dfb7d785be7207                          0.5s
 => [node1 2/9] RUN apt-get update && apt-get install -y     git     gcc     build-essential     libffi-dev     libssl-dev     && rm -rf /var/lib/ap  32.8s
 => [node1 3/9] WORKDIR /opt                                                                                        0.1s
 => [node3 4/9] RUN git clone https://github.com/zhebrak/raftos.git                                                 1.7s
 => [node2 5/9] WORKDIR /opt/raftos                                                                                 0.1s
 => [node2 6/9] RUN sed -i 's/cryptography==1.5.3/cryptography==3.3.2/' requirements.txt                            0.5s
 => [node2 7/9] RUN pip install --no-cache-dir .                                                                    9.7s
 => [node2 8/9] WORKDIR /app                                                                                        0.1s
 => [node2 9/9] COPY node.py .                                                                                      0.1s
 => [node1] exporting to image                                                                                      7.5s
 => => exporting layers                                                                                             4.1s
 => => exporting manifest sha256:50c433e2a78a110770488421fd3a474275cf1d17a6880734b050b16c7299e43e                   0.1s
```

```
 => [node3] resolving provenance for metadata file                                                    0.1s
[+] up 7/7
 ✔Image raft-docker-lab-node3     Built                                                               124.1s
 ✔Image raft-docker-lab-node1     Built                                                               124.1s
 ✔Image raft-docker-lab-node2     Built                                                               124.1s
 ✔Network raft-docker-lab_raftnet Created                                                              0.2s
 ✔Container node3                 Created                                                              1.3s
 ✔Container node2                 Created                                                              1.3s
 ✔Container node1                 Created                                                              1.3s
Attaching to node1, node2, node3
node3  | Starting Node 3 at 172.28.0.4:8003
node3  | [Node 3] Waiting for leader election...
node1  | Starting Node 1 at 172.28.0.2:8001
node1  | [Node 1] Waiting for leader election...
node2  | Starting Node 2 at 172.28.0.3:8002
node2  | [Node 2] Waiting for leader election...
node3  | [Node 3] Leader changed: 172.28.0.3:8002
node3  | [FOLLOWER Node 3] Current leader: 172.28.0.3:8002
node1  | [Node 1] Leader changed: 172.28.0.3:8002
node1  | [FOLLOWER Node 1] Current leader: 172.28.0.3:8002
node2  | [Node 2] Leader changed: 172.28.0.3:8002
node2  | [LEADER Node 2] I am the leader
node3  | [FOLLOWER Node 3] Current leader: 172.28.0.3:8002
node1  | [FOLLOWER Node 1] Current leader: 172.28.0.3:8002
node2  | [LEADER Node 2] I am the leader
node3  | [FOLLOWER Node 3] Current leader: 172.28.0.3:8002
node1  | [FOLLOWER Node 1] Current leader: 172.28.0.3:8002
node2  | [LEADER Node 2] I am the leader
node3  | [FOLLOWER Node 3] Current leader: 172.28.0.3:8002
node1  | [FOLLOWER Node 1] Current leader: 172.28.0.3:8002
node2  | [LEADER Node 2] I am the leader
node3  | [FOLLOWER Node 3] Current leader: 172.28.0.3:8002
node1  | [FOLLOWER Node 1] Current leader: 172.28.0.3:8002
node2  | [LEADER Node 2] I am the leader
node3  | [FOLLOWER Node 3] Current leader: 172.28.0.3:8002
node1  | [FOLLOWER Node 1] Current leader: 172.28.0.3:8002
node2  | [LEADER Node 2] I am the leader
node3  | [FOLLOWER Node 3] Current leader: 172.28.0.3:8002
```

7. Leader Election

Observed logs using docker logs. One node became leader, others followers.

```
ownloads': File exists
PES1UG23AM343:~$ sudo docker logs -f node1
[sudo] password for vardha:
Starting Node 1 at 172.28.0.2:8001
[Node 1] Waiting for leader election...
[Node 1] Leader changed: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
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[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
```

```
PES1UG23AM343:~$ sudo docker logs -f node2
[sudo] password for vardha:
Starting Node 2 at 172.28.0.3:8002
[Node 2] Waiting for leader election...
[Node 2] Leader changed: 172.28.0.3:8002
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
```

```
PES1UG23AM343:~$ sudo docker logs -f node3
[sudo] password for vardha:
Starting Node 3 at 172.28.0.4:8003
[Node 3] Waiting for leader election...
[Node 3] Leader changed: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
```

8. Running Containers

Command:

```
sudo docker ps
```

```
PES1UG23AM343:~$ sudo docker ps
[sudo] password for vardha:
CONTAINER ID   IMAGE                    COMMAND
   CREATED         STATUS         PORTS
                                       NAMES
f062cd6a0eb8   raft-docker-lab-node1   "python node.py 1"
   3 minutes ago   Up 3 minutes
                                       node1
a78eaf575d6c   raft-docker-lab-node2   "python node.py 2"
   3 minutes ago   Up 3 minutes
                                       node2
8c9d1b8a5f31   raft-docker-lab-node3   "python node.py 3"
   3 minutes ago   Up 3 minutes
                                       node3
9602e26335e4   minio/minio             "/usr/bin/docker-ent…"
   3 days ago      Up 38 hours    0.0.0.0:9000-9001->9000-900
1/tcp, [::]:9000-9001->9000-9001/tcp   pes1ug23am343_lab7-min
io-1
```

9. Docker Network

Command:

```
sudo docker network inspect raft-docker-lab_raftnet
```

```
PES1UG23AM343:~$ sudo docker network ls
NETWORK ID     NAME                          DRIVER    SCOPE
73807e8dfd9c   bridge                        bridge    local
cdb2fd7ff451   docker_lab_advanced_chat-net  bridge    local
094f52cb92d5   host                          host      local
ff78703f938f   minikube                      bridge    local
eae57f6c6531   none                          null      local
d02d50fc6935   pes1ug23am343_lab7_default    bridge    local
4b2a018ac35a   raft-docker-lab_raftnet       bridge    local
```

```
PES1UG23AM343:~$ sudo docker network inspect raft-docker-lab_
raftnet
[
    {
        "Name": "raft-docker-lab_raftnet",
        "Id": "4b2a018ac35a5014c48c44d1074159dc1382d149de9c54
5a127e17c3bc2886a9",
        "Created": "2026-03-27T06:27:13.431702682Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": null,
            "Config": [
                {
                    "Subnet": "172.28.0.0/16",
                    "Gateway": "172.28.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {
```

```
                "Name": "node1",
                "EndpointID": "502b75bacf1490b01c818750474e
1b08f7c935a12c517a4257e07814bd7401",
                "MacAddress": "42:3c:c7:11:16:39",
                "IPv4Address": "172.28.0.2/16",
                "IPv6Address": ""
            }
        }
    }
]
PES1UG23AM343:~$
```

10. Leader Failure

Stopped node2. New leader elected automatically.

```
PES1UG23AM343:~$ sudo docker stop node2
node2
PES1UG23AM343:~$
```

```
node2  | [LEADER Node 2] I am the leader
node2 exited with code 137
node3  | [Node 3] Leader changed: 172.28.0.4:8003
node3  | [LEADER Node 3] I am the leader
node3  | [LEADER Node 3] I am the leader
node3  | [LEADER Node 3] I am the leader
node3  | [LEADER Node 3] I am the leader
node3  | [LEADER Node 3] I am the leader
node3  | [LEADER Node 3] I am the leader
node3  | [LEADER Node 3] I am the leader
node3  | [LEADER Node 3] I am the leader
node3  | [LEADER Node 3] I am the leader
node3  | [LEADER Node 3] I am the leader
```

```
[Node 3] Leader changed: 172.28.0.4:8003
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
```

11. Node Recovery

Restarted node2. It rejoined as follower.

```
^CPES1UG23AM343:~sudo docker start node2e2
node2
```

```
Starting Node 2 at 172.28.0.3:8002
[Node 2] Waiting for leader election...
[Node 2] Leader changed: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
```

12. Result

System successfully demonstrated leader election and fault tolerance.

13. Conclusion

Raft ensured consistency and reliability through leader election and recovery.